INCH-POUND

A-A-59165
16 April 2003

PRODUCT DESCRIPTION
DETECTOR, LEAK, HELIUM MASS SPECTROMETER

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers a helium mass spectrometer leak detector. The purpose of the leak detector is to provide the location and measurement of leaks in hermetically sealed vessels, electronic components, or any air/liquid evacuated or pressurized unit.

2. SALIENT CHARACTERISTICS

2.1 General Description. The leak detector shall be portable and weigh less than 75 lbs. (34 kg.).

2.2 Input Voltage Source. The leak detector shall operate at 110 VAC 60 Hz and/or 220 VAC 50 Hz. However, if the solicitation indicates destination(s) outside the U. S., then the offered item(s) shall have the proper power requirements and electric plugs or adapters for the destinations indicated.

2.3 Sensitivity. The leak detector shall have a sensitivity rate of 1×10^{-10} atm. cc/sec (helium) or greater.

2.4 Response Time. The leak detector shall have a response time of less than 2 sec., in accordance with AVS Standard 2.1, Section 3.9.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: WR-ALC/LEEG, 295 Byron street, Robins AFB, GA 31098-1611.

AMSC N/A FSC 6635

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2.5 Test Port Adapter Kit. Test port adapter kit shall contain the couplings for the following sizes of O. D. tubing: $^{1}/_{16}$, $^{3}/_{32}$, $^{1}/_{8}$, $^{3}/_{16}$, ¼, $^{5}/_{16}$, $^{3}/_{8}$, ¾, $^{7}/_{8}$, and 1 inch.

2.6 Roughing Pump. Roughing pump shall be rated 7 cfm. or more.

2.7 Leak Detector Capability/Hardware. Leak detector shall meet it's specified performance characteristics when subjected to the following tests. All equipment required for each test shall be supplied with the leak detector, unless otherwise indicated. Helium source is not required.

2.7.1 ASTM E 493. Standard Test Methods for Leaks Using the Mass Spectrometer Leak Detector in the Inside-Out Testing Mode. Pressure vessel(s) shall be available, but not supplied with the leak detector.

2.7.2 ASTM E 498. Standard Test Methods for Leaks Using the Mass Spectrometer Leak Detector or Residual Gas Analyzer in the Tracer Probe Mode. The helium spray probe kit shall have 8 feet, minimum, of hose. The jet probe(s) shall have a means of controlling the helium flow.

2.7.3 ASTM E 499. Standard Test Methods for Leaks Using the Mass Spectrometer Leak Detector in the Detector Probe Mode. The sniffer probe(s) kit shall have a means of controlling the helium flow.

2.7.4 ASTM E 1603. Standard Test Methods for Leakage Measurement Using the Mass Spectrometer Leak Detector or Residual Gas Analyzer in the Hood Mode. Bell jar(s) and base plates shall be available, but not supplied with the leak detector.

2.8 Test Port Conversion Adapter. One each test port conversion adapter, from 1 1/8 inch tabulation compression fitting to a NW 25 and NW 40 flange, shall be provided.

2.9 Calibration Leak. Calibration leak of 1×10^{-7} std. cc/sec. shall be provided. The calibration leak can be built into the leak detector or external. It shall be traceable to the National Institute of Standards and Technology (NIST).

2.10 Spare Parts and Supplies.

2.10.1 "O" rings and Gaskets. A complete set of "O" rings and gaskets shall be included for the leak detector, test ports, spectrometer tube, and valves, including backfill sections.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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4. QUALITY ACCURANCE PROVISIONS

4.1 Product Conformance. The products provided shall conform to the producer's own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance. Minor changes to the commercial product, with respect to ruggedizing and environmental suitability and protection shall be permitted, subject to the approval of the procurement office, to satisfy the requirements of this requirement.

4.2 Installation and Demonstration. If required by the end user, the contractor shall be responsible for user orientation of the equipment at the designated Air Force facility assuring that the equipment is ready for use and answer any user's concerns about the equipment. Installation and demonstration is projected to take one day. The cost for user orientation shall be at no additional cost to the government.

4.3 Market Acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this requirement.

4.3.1 The manufacturer of the item shall have been producing a product meeting these requirements for at least 5 years.

4.3.2 The manufacturer of the item shall have sold at least 100 units meeting these requirements to the government and/or the commercial market within the last 2 years.

4.4 Alternative offers. A current, complete instrument with all items required for use, shall be offered/provided. If an alternative/replacement item is offered, descriptive literature (commercial catalogs, specification sheets, etc.) shall be submitted with the quote, for review and evaluation by the cognizant engineer who will coordinate qualifying the instrument with the end user (information that is proprietary and should not be released outside the government should be marked with a proprietary statement.) The following information shall be provided: (1) Part Number. Part number and manufacturer of the offered leak detector with a list of items included. (2) Descriptive Literature. Commercial literature (specification sheet) on the offered item. (3) Minor changes. Description of any minor changes to the commercial item description that will be incorporated to meet the CID requirements (examples; sensitivity, tuning and calibration procedures, etc.) (4) Warranty. Reference 3.5.2. (5) Market acceptability. Reference 3.3. If the instrument has not been purchased previously by the government and meets the evaluation process, the instrument shall be subjected to functional testing following prior to final approval by the government. The CID is intended to describe a proven commercial item and does not identify all requirements. Therefore, the government reserves the right to reject an item for characteristics that are not stated in the CID but that are required for acceptance form, fit, and function or suitability for use.

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4.5 Serviceability, Reliability, and Quality of Materials.

4.5.1 One copy of the commercial operation and maintenance manual(s) shall be shipped with the equipment. One additional copy of the operation and maintenance manual shall be shipped to the cognizant engineer at no additional cost to the government. The address for the cognizant engineer is found in 4.5.2.

4.5.2 All supplied items shall carry a standard commercial warranty or a one year warranty on parts and labor which ever is longer. Information on warranty duration and expiration date, conditions and point of contact shall be shipped with each unit. One additional copy of the warranty information shall be shipped to the cognizant engineer at no additional cost to the government. The address for the cognizant engineer is; FSC 6635 Engineer, WR-ALC/LEEG, 295 Byron Street, Robins AFB, GA 31098-1611.

5. PACKAGING. The packing/packaging for shipment to the Government shall be in accordance with contract or purchase order specifications. If specifications are not specified, then the package shall be standard commercial packaging. Additionally, each shipping container shall be marked with the following information or as specified by the contact or purchase order:

Name of equipment
Manufacturer
Model/part number
Contract and/or purchase order number
NSN
Ship to address
Return address

6. NOTES.

6.1 Source of Documents.

6.1.1 ASTM Document. A copy of ASTM documents can be obtained from ASTM 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-5959.

6.1.2 AVS Standards. A copy of AVS standards can be obtained from American Vacuum Society, 120 Wall Street, 32nd Floor, New York, N. Y. 10005.

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6.2 Suggested Sources of Supply.

Varian Inc., Model 979 System consisting of the following items and other items required to meet the CID requirements. Recommended changes to the configuration, to improve performance, ease of use, and/or reduce initial or operating costs, are welcomed. The helium mass spectrometer leak detector shall be shipped under NSN to be provided by the contracting officer. It shall not be shipped under NSN 6635-00-349-8728.

VARIAN INC DBA VARIAN VACUUM TECH. 121 HARTWELL AVE LEXINGTON MA 02421	Catalog #	Description
	979	979 portable leak detector w/turbo version
	K0167301	Helium spray probe kit
	K9565301	Power probe sniff accessory 1 1/8" test port fitting
	670029096	Spectrometer tube cleaning kit
	86428801	Test port adapter kit
	L8741301	Test port conversion kit, 1 1/8" test port to NW25
	L8741302	Test port conversion kit, 1 1/8"test port to NW 40

MILITARY INEREST:

Custodian:
Air Force - 99

Preparing Activity:
Air Force – 84

Agent:
Air Force - 99

Project No. 6635-0204

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